

Cost-Profit Optimization Model for Highway Rest Area EV Charging Station DES

C221017 Hwisu Kim

June 17, 2026

1 Introduction

This report proposes a Discrete Event Simulation (DES) model to analyze the operational efficiency of an EV charging station at a hypothetical highway rest area and to derive an optimal operational policy. While the initial proposal established the basic framework using limited server capacity and simple First-Come, First-Served (FCFS) assumptions, this final report addresses those limitations. It incorporates realistic constraints such as customer movement time within the rest area, a priority queueing system, and cost-profit optimization logic.

2 Changes from Previous Model and New Assumptions

The major changes reflecting the feedback can be summarized as follows:

- **Advanced Customer Behavior-Based Decision Making**
 - **Charger Selection Routing:** Divided into two types: pre-determined assignment vs. selection after checking the current queue length.
 - **State-Based Behavior (Policy → Behavior):** Introduced 'snack purchase probability' and 'priority ticket' purchasing based on expected waiting time, moving beyond simple system control.
- **Realistic Physical Environment and Constraints**

- **Charger Types and Infrastructure Expansion:** Expanded from 2 chargers (1 fast, 1 slow) to a total of 8 (4 ultra-fast, 4 fast), reflecting the reality of highway rest area infrastructure.
- **Battery Capacity Integration:** Fixed charging time by vehicle type → Charging time calculated proportionally to the remaining battery level upon entry.
- **Queue Space Limit Adjustments:** Finite capacity ($K = 20$), but allows partial infinite queuing (∞) for exception cases.
- **Statistical Modeling and Objective Function Specification**
 - **Cost-Profit Optimization:** $\max [(\text{Charging} + \text{Snack} + \text{Priority Ticket Revenue}) - (\text{CAI} + \text{Queueing})]$
 - **Queue Discipline:** Simple FCFS → Profit-generating Priority Queue.
 - **Kendall Notation:** $M(t)/G/2/K/FCFS \rightarrow M(t)/G/8/K/\text{Priority}$

The model now reflects the psychology of customer charger selection, infrastructure expansion, battery-proportional charging times, and cost-profit optimization. It is assumed that vehicles with a battery level of 10% or less (at risk of discharge) will purchase priority tickets with a 100% probability. Furthermore, since these vehicles cannot reach the next rest area, they are exceptionally allowed to enter the queue even if the system capacity (K) is full.

2.1 Arrival Process and Charging Time

The final charging time (Service Time) of an individual vehicle entering the system is not directly drawn from a single probability distribution. Instead, it follows a general probability distribution (G) determined by a combination of three independent variables: Vehicle Type, Initial Battery State, and Charger Performance (Service Rate). The specific step-by-step modeling process is as follows:

Step 1: Arriving Vehicle PMF Setup

The battery capacity by vehicle type is defined as a discrete random variable X :

$$P(X = x_j) = p_j$$

- $x_1 = 50$ (Compact Car) : $P(X = 50) = 0.20$
- $x_2 = 77$ (Sedan) : $P(X = 77) = 0.60$
- $x_3 = 100$ (Truck) : $P(X = 100) = 0.20$

Step 2: Determining Vehicle Type (V) using Inverse Transform Method

The vehicle type is determined using the Inverse Transform Method for discrete probability distributions. After generating a random number $U_1 \sim \text{Uniform}(0, 1)$, the vehicle type V is mapped according to the Cumulative Distribution Function (CDF) intervals:

$$V = \begin{cases} \text{Compact,} & 0 \leq U_1 < p_1 \\ \text{Sedan,} & p_1 \leq U_1 < p_1 + p_2 \\ \text{Truck,} & p_1 + p_2 \leq U_1 \leq 1 \end{cases} \quad (1)$$

Based on the determined vehicle type V , the total battery capacity constant Cap_V for the vehicle is set ($Cap_V \in \{50, 77, 100\}$).

Step 3: Determining Initial Battery State ($Battery_{first}$) using Asymmetric Distribution

To reflect the realistic customer behavior pattern where vehicles with lower battery levels have a higher probability of entering the charging station, the initial battery state $Battery_{first}$ is modeled using a right-skewed continuous probability distribution. This study applies a Beta Distribution within the range of $[0, 1]$.

Since drivers with low batteries are much more likely to visit the station, α is set to 2 and β to 5, ensuring that a battery level around 20% appears most frequently.

$$Battery_{first} \sim \text{Beta}(2, 5) \quad (2)$$

If the generated random value is $Battery_{first} \leq 0.1$ (battery 10% or less), the vehicle is classified as a 'discharge risk vehicle' and exhibits the behavior of purchasing a priority ticket with a 100% probability. The target charging state is uniformly set to $Battery_{target} = 0.8$ (80%) reflecting battery protection protocols.

Step 4: Calculating Final Charging Time (S)

Let $Power_j$ be the power supply rate and $EFFICIENCY_j$ be the charging efficiency according to the chosen charger type $j \in \{\text{Ultra-fast}(1), \text{Fast}(2)\}$. The final charging time S is dynamically calculated by the following formula:

$$S = \frac{Cap_V \times (0.8 - Battery_{first})}{Power_j \times EFFICIENCY_j} \quad (3)$$

2.2 Movement Time within the Rest Area

As in the previous proposal, physical movement time from the rest area entrance to parking is excluded from the calculation. Instead, the simulation arrival event (t_A) is redefined not as 'passing the rest area entrance' but as 'the moment of reaching the parking lot (charging waiting area)'.

2.3 Priority Tickets and Queue Discipline

The concept of a priority ticket was introduced as a non-preemptive priority queue. It does not push out a vehicle currently charging, but rather places the ticket holder next in line once a charger becomes available. Priority tickets can only be used for ultra-fast chargers.

Since customers do not know the number of waiting vehicles before entering the rest area, they are divided into those who purchase in advance and those who decide after seeing the queue length on-site. (It is assumed all customers can estimate waiting times).

The price of the priority ticket is set at 3,500 KRW.

- **Calculating Theoretical Average Required Charge ($E[\Delta Q]$):**

To determine the price, we first calculate the expected value of the required charge.

$$E[\Delta Q] = E[Cap_V] \times (Battery_{target} - E[Battery_{first}])$$

$$E[\Delta Q] = 76.2 \times \left(0.8 - \frac{2}{7}\right) \approx 39.19 \text{ kWh}$$

- **Calculating Average Time Saved (ΔW):**

Based on the expected average required charge (approx. 39.19 kWh), we derive the time difference between using a fast charger (100kW)

and an ultra-fast charger (350kW).

$$\Delta W = \left(\frac{39.19}{100 \times 0.90} \right) - \left(\frac{39.19}{350 \times 0.88} \right) \approx 0.31 \text{ hours} \quad (4)$$

- **Converting Time to Economic Value and Final Pricing:**

By multiplying the saved time (ΔW) by the expected minimum wage for 2026 (10,320 KRW/hr), the customer's temporal opportunity cost is converted into monetary value.

$$\text{Opportunity Cost} = 0.31 \text{ hrs} \times 10,320 \text{ KRW/hr} \approx 3,199 \text{ KRW} \quad (5)$$

In this simulation model, a price slightly above the calculated opportunity cost, **3,500 KRW**, is applied for the priority ticket.

- **Priority Ticket Purchasing Groups**

- **Pre-purchase Group:** Customers who pre-pay for a priority ticket via an app in a state of information asymmetry (without knowing the congestion) before entering the rest area. Occurring at a fixed proportion (20%) of total customers via a Bernoulli distribution, they join the VIP queue immediately upon arrival regardless of the queue length.
- **On-site Purchase Group:** Customers who arrive without pre-purchasing and decide on-site. The purchase logic branches into two based on the remaining battery state ($Battery_{first}$) upon entry:
 - * **Discharge Risk Customers:** If remaining battery is 10% or less, they immediately purchase a ticket with 100% probability to prevent discharge and enter the VIP queue, regardless of queue length or Expected Waiting Index (EWI). (They cannot leave the queue even if capacity is full).
 - * **Normal Waiting Customers:** If battery levels are sufficient, they visually check the ultra-fast and fast queue lengths (n) and EWI before deciding. They probabilistically purchase tickets based on a Stepwise Function where purchase probability increases as expected waiting time increases.

Modeling the Priority Ticket Purchase Decision

A customer's decision to purchase is sequentially determined by 'pre-purchase status' before entry and 'on-site status' after entry. Let event A be pre-purchasing and event B be on-site purchasing.

– **Step 1: Pre-purchase Decision (Bernoulli Trial)**

Vehicles pre-purchase with probability $p_1 = 0.2$.

$$P(A) = 0.2$$

Generate $U_1 \sim \text{Uniform}(0, 1)$; if $U_1 \leq 0.2$, pre-purchase (VIP queue entry) is confirmed.

– **Step 2: On-site Decision (Nested Conditional Bernoulli)**

For the remaining 80% (A^c), the on-site purchase probability $P(B|A^c)$ depends on $Battery_{first}$ and the **Minimum Expected Waiting Time** $t = \min(EWI_{fast}, EWI_{ultra})$.

$$P(B|A^c) = \begin{cases} 1.0 & (Battery_{first} \leq 0.1 \quad : \text{Discharge Risk}) \\ p_2(t) & (Battery_{first} > 0.1 \quad : \text{Normal Customer}) \end{cases} \quad (6)$$

Where the purchase probability function $p_2(t)$ follows a stepwise function:

$$p_2(t) = \begin{cases} 0.0 & (t < 10 \text{ min}) \\ 0.1 & (10 \leq t < 30 \text{ min}) \\ 0.25 & (t \geq 30 \text{ min}) \end{cases} \quad (7)$$

- **Simulation Algorithm Implementation:** Among vehicles without pre-purchase ($U_1 > 0.2$), those with $Battery_{first} \leq 0.1$ enter the VIP queue unconditionally. Normal vehicles ($Battery_{first} > 0.1$) generate $U_2 \sim \text{Uniform}(0, 1)$; if $U_2 \leq p_2(t)$, they join the VIP queue, otherwise they join the regular queue.

• **Non-purchasing Group**

- General customers who do not purchase a priority ticket calculate the Expected Waiting Time (EWI) and dynamically route to the

faster option between ultra-fast and fast chargers.

The formula for calculating EWI is as follows:

Expected Waiting Index (EWI)

$$EWI = \frac{n}{c \times \mu} \quad (8)$$

Variable Definitions

- n : Current number of waiting vehicles in the respective queue
- c : Number of operating servers (chargers)
- μ : Service rate per unit time per charger

To calculate EWI, the service rate is required. Reflecting realistic charging efficiencies, the service rates for fast and ultra-fast are:

- **Theoretical Service Rate (μ) Derivation**

- **Normal Fast (100kW, Efficiency = 0.90)**

$$E[S_{fast}] = \frac{39.19}{100 \times 0.90} \approx 0.435\text{h} \quad \Rightarrow \quad \mu_{fast} \approx 2.30 \text{ vehicles/hr}$$

- **Ultra-fast (350kW, Efficiency = 0.88)**

$$E[S_{ultra}] = \frac{39.19}{350 \times 0.88} \approx 0.127\text{h} \quad \Rightarrow \quad \mu_{ultra} \approx 7.86 \text{ vehicles/hr}$$

EWI utilizing Service Rate and Number of Chargers (Baseline 4, 4)

- **Ultra-fast (VIP) Queue EWI (EWI_{ultra}):**

$$EWI_{ultra} = \frac{n_{ultra}}{c_{ultra} \times \mu_{ultra}} = \frac{n_{ultra}}{4 \times 7.86}$$

- **Normal Fast Queue EWI (EWI_{fast}):**

$$EWI_{fast} = \frac{n_{fast}}{c_{fast} \times \mu_{fast}} = \frac{n_{fast}}{4 \times 2.30}$$

2.4 Customer Loss Modeling (Balking and Blocking)

To reflect realistic constraints of the rest area, two loss logics are applied.

- **Voluntary Abandonment (Balking):**

If the minimum expected waiting time ($\min(EWI_{fast}, EWI_{ultra})$) exceeds a customer's tolerance time, they balk (abandon the queue). Tolerance Time (T_{tol}) follows a normal distribution with mean 1.5 hrs and SD 0.4 hrs.

$$T_{tol} \sim N(1.5, 0.4^2) \quad (\text{Min bounded to } 0.5)$$

(Priority ticket holders do not balk).

- **Involuntary Abandonment (Blocking):**

Due to physical parking space limits ($K = 20$), if $N_U + N_F = 20$, new general customers are blocked and lost. (Priority ticket holders can exceptionally bypass this limit).

2.5 F&B Revenue and Purchase Probability Modeling

Secondary revenue (food beverage) generated during waiting and charging is considered. Customer purchase depends on Total Stay Time (W_{total}). Upon departure, revenue is generated based on W_{total} ($Wait + Charge$) using the following probabilities:

- **Menu Items:**

- Light: 2,500 KRW (Beverages, Americano, etc.)
- Medium: 6,000 KRW (Snack sets, Hotdogs, etc.)

- **Case 1: $W_{total} < 15$ min** (Very short stay, e.g., zero-wait ultra-fast)

- Restroom use only; low consumption.

- Probability: Light 15% / Medium 5% (None 80%)
- **Case 2:** $15 \text{ min} \leq W_{total} < 40 \text{ min}$ (Normal fast charging or slight waiting)
 - Customers browse stores and consume takeout snacks.
 - Probability: Light 40% / Medium 20% (None 40%)
- **Case 3:** $W_{total} \geq 40 \text{ min}$ (Trapped in long queues)
 - Active consumption of high-margin sets to alleviate boredom.
 - Probability: Light 30% / Medium 60% (None 10%)

3 Simulation Mechanism: Variables and Initialization

System Variables

- **Time Variable:** t
- **System State (SS):** (N_U, N_F)
 - N_U : Total vehicles in Ultra-fast system (Queue + Charging)
 - N_F : Total vehicles in Fast system (Queue + Charging)
- **Counter Variables**
 - N_A : Cumulative arrivals
 - C_{ultra}, C_{fast} : Completed services
 - $Revenue$: Cumulative total revenue
 - $N_{balking}$: Cumulative voluntary losses
 - $N_{blocking}$: Cumulative involuntary losses
- **Event List**
 - t_A : Next arrival time
 - E_{ultra} : Set of scheduled completion times for ultra-fast chargers.
($t_{D_ultra} = \min(E_{ultra})$)
 - E_{fast} : Set of scheduled completion times for fast chargers. ($t_{D_fast} = \min(E_{fast})$)

Initialization

- **Variables:** Set $t = N_A = C_{ultra} = C_{fast} = Revenue = 0$
- **Loss Variables:** Set $N_{balking} = 0, N_{blocking} = 0$
- **State:** Set $SS = (0, 0)$
- **Event Scheduling:** Generate first arrival T_0 , Set $t_A = T_0, t_{D_ultra} = \infty, t_{D_fast} = \infty$

Recursive Loop (Case 1: Arrival)

- **Condition:** $t_A = \min(t_A, t_{D_ultra}, t_{D_fast})$
- **Execution:**
 - Update time $t = t_A$, increment N_A .
 - Schedule next arrival $t_A = t + Y_A, N_A = N_A + 1$.
 - Assign attributes $(Cap_V, Battery_{first})$.
 - Determine VIP Ticket (Pre-purchase / On-site logic).
 - If no ticket: evaluate Balking and Blocking logic. Update $N_{balking}$ or $N_{blocking}$ and terminate case if lost.
 - If admitted:
 - * If VIP Ticket: Join Ultra-fast queue with Priority = -1. Update $SS = (N_U + 1, N_F)$.
 - * If General: Compare EWI_{ultra} and EWI_{fast} , join the shorter queue with Priority = 0. Update $SS = (N_U + 1, N_F)$ or $SS = (N_U, N_F + 1)$ accordingly.
 - Check idle servers for the chosen system. If available, calculate S , start charging, and update E_{ultra} or E_{fast} .

Recursive Loop (Case 2: Ultra-fast Departure)

- **Condition:** $t_{D_ultra} = \min(t_A, t_{D_ultra}, t_{D_fast})$
- **Execution:** Update $t = t_{D_ultra}$. Execute Revenue logic. Increment C_{ultra} . Update $SS = (N_U - 1, N_F)$. Remove event from E_{ultra} . If $N_U \geq c_{ultra}$ (where N_U is the updated state), pop top priority vehicle from queue, start charging, and update E_{ultra} .

Recursive Loop (Case 3: Fast Departure)

- **Condition:** $t_{D_fast} = \min(t_A, t_{D_ultra}, t_{D_fast})$
- **Execution:** Update $t = t_{D_fast}$. Execute Revenue logic. Increment C_{fast} . Update $SS = (N_U, N_F - 1)$. Remove event from E_{fast} . If $N_F \geq c_{fast}$ (where N_F is the updated state), pop top priority vehicle from queue, start charging, and update E_{fast} .

4 System Performance Measures and Objective Function

4.1 Performance Measures

- **System Average Waiting Time ($E[W_q]$):** Overall average time spent in queue before charging begins.
- **System Average Stay Time ($E[W_{total}]$):** $E[W_{total}] = E[W_q] + E[S]$.
- **Charger Utilization (ρ):** ρ_{ultra} and ρ_{fast} calculated separately to evaluate load balance.
- **Probability of Loss (P_{Loss}):** Ratio of lost customers to total arrivals,
$$P_{Loss} = \frac{N_{loss}}{N_A} = \frac{N_{balking} + N_{blocking}}{N_A}.$$

4.2 Cost-Profit Objective Function

The objective is to maximize the net profit (Z) by minimizing cost and maximizing revenue:

$$\max Z = \text{Total Revenue (TR)} - \text{Total Cost (TC)} \quad (9)$$

- **TR:** Customer charging fees + Priority ticket sales + F&B (snack) sales.
- **TC:** CAPEX (Monthly depreciation of chargers) + OPEX (Fixed maintenance costs).

5 Simulation Results and Scenario Analysis

5.1 Operational Metric Results

Operational Metric	Alt 1 (4, 4)	Alt 2 (5, 4)	Alt 3 (4, 5)
Average Waiting Time, $E[W_q]$ (min)	0.92	0.25	0.84
Average Stay Time, $E[W_{total}]$ (min)	11.12	9.99	11.46
Customer Loss Rate, P_{Loss} (%)	0.00138%	0.00004%	0.0090%

Table 1: Comparison of Operational Metrics across Alternatives

In terms of operational efficiency, **Alternative 2** was the optimal solution. The addition of one ultra-fast charger drastically reduced the average waiting time by **73%** and the customer loss rate by **97.1%** compared to the baseline.

5.2 Financial Metric Results

Financial Metric	Alt 1 (4, 4)	Alt 2 (5, 4)	Alt 3 (4, 5)
Total Revenue, TR (KRW)	377,843,340	377,170,022	377,173,645
Total Cost, TC (KRW)	5,015,200	9,301,533	6,399,333
Net Profit, Z (KRW)	372,828,140	367,868,489	370,774,312

Table 2: Comparison of Financial Metrics across Alternatives

Despite the operational brilliance of Alternative 2, the economic optimization yielded a contradictory result. The steep CAPEX and OPEX associated with infrastructure expansion eclipsed the marginal revenue gained from capturing lost customers. Thus, **Alternative 1** yielded the maximum net profit (Z).

6 Conclusion and Future Work

6.1 Conclusion

- **Optimal Scenario: Alternative 1** (Baseline: 4 Ultra-fast, 4 Fast chargers).
- **Infrastructure Adequacy:** Under the currently modeled arrival frequency, the existing 8 chargers were sufficient to prevent severe bottlenecks.

- **Economic Rationale:** The increase in fixed infrastructure investment (depreciation) and maintenance costs outweighed the expected revenue from increased charging volume, snack sales, and priority tickets. Therefore, maintaining the current infrastructure without expansion is the most economically sound decision.

6.2 Academic Contribution

This study’s primary contribution lies in successfully transitioning from passive FCFS models to dynamic, behavior-driven queueing networks. By accurately translating psychological thresholds (Balking) and value-based actions (Priority Tickets) into quantitative probabilities, the simulation successfully captures and reflects realistic customer decision-making processes.

6.3 Limitations and Future Research

Despite its contributions, this study presents several limitations. First, assumptions regarding flat utility pricing and assumed POS (Point of Sale) margins constrain the depth of the economic analysis. More critically, the current model assumes a **Homogeneous Poisson Process** with a constant arrival rate (λ), which evenly distributed the incoming traffic across the simulation period and prevented extreme bottlenecking. This lack of severe traffic jams limited the visibility of priority ticket profitability.

Future research should address these limitations by applying a **Non-homogeneous Poisson Process (NHPP)** with a time-varying arrival rate ($\lambda(t)$) to accurately capture realistic traffic surges during specific periods (e.g., peak commuting hours, national holidays). Furthermore, incorporating empirical POS datasets, temporal traffic volumes, and digital twin stress-testing will provide more granular guidelines for real-world CAPEX investments and optimal infrastructure planning.